GeoAI foundation models - Will they have the same fate as upper ontologies in GIScience?

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Abstract

From the 1990s to the 2010s, researchers in the field of GIScience were trying to formulate an upper ontology to formalize geographic information on the uppermost level possible. While a range of different approaches stemming from physical or linguistic considerations have been introduced, no final upper ontology could be formulated and interest in the topic seemingly disappeared. With developments in AI reaching a new peak every year, research in finding a foundation model in GeoAI grows. Over the past years, several potential attempts at creating such models have been made. While it remains unclear how such a foundation model should look like in detail, it is clear that the multitude of possible tasks and the multi-modal nature of spatial data pose great problems. Though AI is at the center of research in academia and business, it remains unclear whether the same will hold true for GeoAI as research is tightly tied to monetary interests. Further, a lack of an upper ontology might hinder developments and lead to the research being abandoned, as well.

1 Introduction

Finding a common ground in the representation of information is essential in science. Without shared standards and frameworks, meaningful collaboration across countries, institutions, and linguistic or cultural backgrounds becomes extremely difficult. This is particularly true in the field of Geographic Information Science (GIScience), where researchers study objects and phenomena that are inherently tied to physical space, cultural context, and often ambiguous or fluid boundaries. For effective scientific communication and collaborative work, agreements must be reached not only on how to describe and analyze spatial phenomena, but also on how to categorize and conceptualize them. As Fisher and Wood (1998) mention, such seemingly simple questions as "What is a mountain?" lead to greatly different answers, depending on the cultural background and geographical circumstances, while making distinctions between similar objects is difficult and transitions often fuzzy. Mark et al. (1999) found, that even within small communities, such differences in categorizing geographic entities remain. In order to combat these difficulties, the University Consortium for Geographic Information Science (UCGIS) launched a research initiative in 1998. Researchers in the field were animated to work towards a shared upper ontology in the field of GIScience. While several proposals have been made, as of 2025 no definite one exists, and the topic seemingly has been put to rest.

At the same time, the use of artificial intelligence (AI) has been increasing rapidly. In 2024, more than 13% of companies in the European Union (EU) were using AI in some way (eurostat, 2025), with large businesses exceeding 40%. In a global survey, Gillespie et al. (2025) found 66% of the world's population using AI in 2024. The UN is projecting a $4.8 trillion AI market for the year 2033 (UNCTAD, 2025). Most notably, AI is being used in the form of large language models (LLMs) as conversational tools or chat bots. These models are referred to as foundation models due to their capacity to perform a broad range of tasks without requiring task-specific modifications by the user. However, these models have been found to yield insufficient results when met with tasks related to spatial problems (Yiannakoulias, 2024; Li et al., 2024; Liu et al., 2025), thus showing the importance of creating special foundation models for GIScience and related areas of research. These new developments nevertheless lead to new difficulties. Although LLMs are shown to work for various tasks, the lack of spatial understanding and reasoning poses great difficulties. Not having a universal upper ontology further complicates the matter, as underlying mechanisms of LLMs necessarily need a common framework to fulfill tasks. While the quest for finding an upper ontology in GIScience has been put to rest as of now, the future of GeoAI foundation models is unsure. Will they also end as a comparatively short-lived trend? Or can the development of such models lead to the revival of the search and formulation of an upper ontology?

2 Related Work

2.1 Upper Ontologies

In information science, an upper ontology (UO) is a fundamental, abstract model that conceptualize top-level objects independent of specific domains. Biagetti (2021) describes how the combination of term taxonomies, axioms, and terminological dictionaries can lead to building such UOs and help with information sharing across database management systems and software applications. Niles and Pease (2001) describe such a UO developed by the IEEE Standard Upper Ontology Working Group. Based on lower-level ontologies, the group developed SUMO (Suggested Upper Merged Ontology) as a UO categorizing and merging underlying objects as physical and abstract entities with relationships and action between them. Via connections between different entities, a complex blueprint could be constructed for domain-specific applications. Batres et al. (2007) propose a similar blueprint for the field of plant science, which lead to similar developments in waste, chemical, or computer-aided process engineering. Mascardi et al. (2007) compare different UO schemes in use and how they can be applied to allow for unified information processing. Mascardi et al. (2010) further explain how the use of UOs can help connecting ontologies of lower levels and allow for information processing and work across different sub-disciplines.

2.2 Upper Ontologies in GIScience

In the field of GIScience, the search for an UO has been focused on from the late 1990s to the early 2010s. Claramunt and Thériault (1995) express the necessity to integrate space as known links of events and consequences into an UO. Addressed first by the UCGIS in 1998 as an emerging field and the need of research, a range of contributions have been made. Smith and Mark (2001a) and Mark et al. (2004) find the need to ”build robust, comprehensive, and usable taxonomies" in the field of geographic information and to seek "appropriate representations for geospatial phenomena to match the underlying reality". Fonseca et al. (2006) identify a need of an UO in order to allow for interoperability between different Geographic Information Systems (GIS). While agreements are available in small communities, the development of new tools and access to an increasingly larger amount of data necessarily lead to a loss of meaning across communities. While ontologies exist in small sub-domains, an overarching conceptual model needs to be found. Smith and Mark (2001b) further find the necessity to distinguish between geographical entities as physical and social concepts. While have been investigated, more emphasis has been placed on the social, semantic, and institutional views of geographic space.

2.2.1 Geographic objects as physical entities

Smith and Mark (2001b) present a first idea of approaching the representation of geographic entities as distinct physical objects. Treating objects in space as bona fide objects, they describe them as real entities of universal categories. Based on physical boundaries and features, objects such as mountains are to be described in the same manner universally. In addition, continuous phenomena in time and space, such as temperature or atmospheric activity, are to be treated using field theory. Goodchild et al. (2007) introduce the concept of geo-atoms as geographic primitives similar to physical atoms, representing a single point of space and its related properties. Physical space is thus constructed from a countable number of single geo-atoms or collections of related, close-by groups of geo-atoms with an UO describing their relationships.

2.2.2 Geographic objects as semantic concepts

Smith and Mark (2001b) identify an inherent vagueness and ambiguity in the description of geographic objects that stem from linguistic and semantic ambiguities. Although bona fide considerations are possible in theory, cultural, social, and institutional concepts play a large role in defining entities making them fiat objects. Geographic realities are created by human reasoning and language and show differences in categorization. These differences are inherently incompatible with a possible UO and have to be overcome. To overcome these differences, Mika et al. (2004) introduce semantic interoperability via meanings. Instead of using single word descriptors to describe physical objects, descriptions are connected via an UO. Basic underlying categories of human cognition should be used to create concepts and relations. However, this comes with challenges regarding human perception and related social agreements. It is unavoidable to introduce vagueness, uncertainty, and different levels of granularity into the conceptualization process. Matching different agreements leads to a challenging reasoning process in order to overcome semantic heterogeneity. (Kuhn, 2001, 2005; Kuhn et al., 2007). Sinha and Mark (2010) proposes the use of common-sense categories and terms linked by universal spatial relationships, describing single entities of objects through their related physical entities, thus bridging the gap between bona fide and fiat approaches. Janowicz (2012) proposes an ontology from the bottom up. Instead of building a UO from the top to unite existing ontologies, advances in big data collection should be utilized to restructure concepts from observations.

2.2.3 Geographic objects as information constructs

Couclelis (2010) introduces a third approach to find a UO in GIScience. Instead of approaching geographic objects as physical or linguistic representations, entities could also be treated as discrete observations in space and time. Treating observations as pure information, a meta-ontology could be constructed based on fundamental information science. Specific meaning can be reduced through semantic contraction, leading to discrete information entities by decreasing the level of meaning. Not only would this result in a single descriptor for different objects and relations, but the "hierarchy of discourse" could allow for different semantic levels for different use cases. Rather than detailed descriptions in an UO, distinct types of information constructs could be achieved in decreasing detail.

2.3 GeoAI foundation models

Foundation models for geographic information are a relatively new concept currently drawing significant research interest. There is no consensus on what constitutes a GeoAI foundation model, and definitions vary across the field. Given that GIScience and its applications span a broad range of tasks and data types, multi-modality is central to the development of a foundation model. While human interaction with AI is typically based on written or spoken language, geographic data seldom comes in text form. Instead, it often takes the shape of vector or raster data, as well as images (Wang et al., 2024). Xie et al. (2023), Zhang et al. (2023), and Xu et al. (2025) explore the use of LLMs for spatial data, defining GeoAI foundation models from a perspective of language processing and integration of spatial data into reasoning. Hsu et al. (2024) and Google (2025) focus on AI models for processing remote sensing image data. The challenges posed by the wide variety of tasks make it difficult to arrive at a single, clear definition, as the scope and focus of these models depends heavily on the researchers' expertise and viewpoint. One common ground across these approaches is the use of machine learning technologies on very large datasets. By leveraging spatial reasoning, models should be pretrained to handle a variety of tasks, providing flexible solutions without being tied to any one specific task.

3 The fate of upper ontologies

Although a wide range of contributions have been made to the field, no distinct UO in GIScience could be found or agreed on. While semantic web implementations introduced some level of standardization to geographic information, differences remain and formalization remains difficult, especially when expert knowledge is involved. The standards developed by bodies such as the Open Geospatial Consortium (OGC) or International Organization for Standardization (ISO) are not fully extensible and reusable across all domains involved in GIScience and rapid developments in the field pose difficulties in adapting in the future. Different ontology language editors exist, but are not built with focus on complete interoperability and with sensibility to the specific field. Even though large repositories of formal representations and concepts exist, they are not necessarily interoperable or manipulatable on higher levels. (Claramunt, 2020).

As of 2025, the search for an upper ontology in GIScience has been largely abandoned. Though standardization and advancements of the semantic web allow for better understanding and collaboration, concepts of geographic objects still vary and are not fully harmonized on an upper level. Instead of focusing on such fundamentally scientific issues as finding a UO, GIScience developed further into finding applications as an auxiliary science and creating marketable solutions.

4 The state of foundation models in GeoAI

Research into GeoAI foundation models is still in its early stages, and as such, literature on the topic remains sparse with no definitive models in existence. Xie et al. (2023) pose the fundamental question what a GeoAI is at its core. Applying approaches from general AI research on spatial data ignores problems inherent to geography. Simply using existing or future general foundation models renders the ”Geo”-prefix irrelevant and does not leverage possible potential. Additionally, collecting and annotating large datasets introduces more issues. Annotation is necessarily connected to human bias, ideas, and expectations. Foundation models trained on these datasets would simply imitate human understanding without deeper insights or reasoning. Labeling the data is vastly different from daily conversation and thought about geographic realities and can not fully mirror it as labels are both limited in number and must follow a strict definition. This issue is further emphasized by the nature of data collected from a large range of sources. As mentioned above, ideas about space vary depending on culture or language and can not easily be summarized under the same label. Expert knowledge is necessary to create spatial datasets for foundation models and the heterogeneity in data and phenomena across the globe pose additional difficulties. Generalizations from limited areas can not be made and labeling data from virtually everywhere is hardly possible due to costs and work needed. At the same time, similar phenomena in different areas can lead to conflicting entities of the same label when attempted to be harmonized. Spatial tasking must thus necessarily be built with keeping awareness of heterogeneity in mind. Specific tasks must follow a self-supervised training method to overcome human bias in training data. However, such universal methods have yet to be found. Collecting vast amounts of data requires high levels of cooperative work between science, industry, and the public, as well as precise ground-truthing usable data not a given at this point. Integration of communities across the world and society is necessary to avoid bias in possible models. While LLM tasks can be realized more easily, multi-modality is a great challenge at this point.

Mai et al. (2024) and Xu et al. (2025) highlight the potential of language-based tasks, noting that current state-of-the-art LLMs can handle basic spatial reasoning and answer simple text-based questions. However, more complex spatial tasks, such as forecasting, remain highly inaccurate. Multi-modality continues to be a significant challenge, as different data types cannot be fully integrated or harmonized, preventing the development of true GeoAI foundation models. Different data types can not be fully harmonized and linked, leaving the greatest challenge in building GeoAI foundation models open. Hallucinations from model inference further introduce uncertainty into results.

Wang et al. (2024) compare a range of domain-specific foundation models in different fields of GIScience. Effective on specific tasks, it is questionable whether they can be considered as foundation models for wider areas of research. While they can be used to fulfill these tasks, they remain unaware of space and lack spatial reasoning. Context is lost as text or images are treated without consideration of the geographic or societal environment they appear in. Multi-dimensionality is inherent to spatial data and related processing costs intensive. The resources needed to integrate a wide range of data might exceed possibilities of research institutions and outweigh the necessity of a GeoAI foundation model. Another major challenge is the non-Euclidian nature of space and integration of a large number of different reference systems and projections. Data harmonization becomes necessary but remains difficult. Further, AI technologies assume independence of measurements, violating Tobler's first law of geography. Geographic entities are related on different levels and can not be considered as fully independent, a problem that traditional, non-spatial models do not fully implement. Complex spatial relationships, patterns, and context, such as proximity or connectivity between different geographic features, require a level of spatial reasoning that is yet to be implemented into GeoAI models. Given the changing nature of geographic data, scalability is required to lead to accurate and up-to-date results, further increasing cost and computational resources needed.

5 GeoAI foundation models - Will they have the same fate as upper ontologies in GIScience?

5.1 Overcoming Challenges

The remaining challenges mentioned above currently pose difficulties in realizing a GeoAI foundation model, but could be overcome. Standardization in recording and storing data, as well as the emergence of the semantic web can help structure data in a meaningful manner. Further standards could be formulated to reduce pre-processing needs for larger amounts of data and make it more readily usable in AI. As models for specific tasks and areas of interest are created at an increasing rate, new developments might help achieving multi-modal foundation models. Implementations of connected domain-specific models could be treated as ”arms” of a potential foundation model being activated when needed. As more and more data is being made available via the internet, further spatial information can be gathered. Advanced data mining techniques could allow for this to happen in real time and at lower personnel costs in an automated manner. New developments in semi-supervised or unsupervised learning could further minimize the need of sampling training datasets. Combining existing datasets currently used for very specific tasks and harmonizing them to work in a foundational model could leverage already existing expert knowledge. Further, the use of knowledge graphs could introduce some level of consistency and reasoning into a potential model. Rethinking how data is collected and applying standards in accuracy and ground-truthing could help with data quality in the future and allow for a focus on recording and creating AI-ready data.

5.2 GeoAI foundation models and Upper Ontologies

The most pressing issue for developing a foundational GeoAI model remains the lack of an overarching upper ontology in the field. Due to the nature of AI models reducing complexity of data, it is of great importance to understand, formalize, and discuss phenomena at the lowest possible level. Universal representations must be found not only to allow AI to understand data in a universal fashion, but also for results to be understood by humans and consistent across space and domain. As data is increasingly being collected from publicly available sources or contributed by the public, a human bias automatically enters the data. Different people understand geography and geographic objects differently than others and culture and language play a significant role. The general public does not follow stringent definitions in everyday conversation or when talking about geographic phenomena. It is unavoidable that a general understanding of the topic has to be formalized to bring different voices together in a coherent system. Meaning must be captured and conceptualized for data harmonization and understanding. While - in theory - different meanings could be fed into a model, results might be inconsistent or not applicable in all parts of the world. Further, the amount of data necessary to capture all possible meanings and return results fitting to a task’s or user’s context is computationally unwise. The quest for a GeoAI foundation model could revive the search for upper ontologies. Although earlier research was founded on pure scientific inquiry and for purposes of science, the popularity of AI might give new considerations to the idea. Given the monetary success of foundation models in language processing and interest of companies to introduce spatial reasoning to their models, an upper ontology could be of greater importance than ever. Standard formalizations of spatial phenomena are necessary for models to achieve universal applicability and high performance and reliability. AI having to distill data to information at the lowest possible level might help giving new ideas on how to do this with geographic information. Additionally, new capacities in collecting and processing extremely large amounts of data from around the world might help reaching a deeper understanding of how geography is understood and experienced and give insight into common denominators. At the same time, developments could lead in the opposite direction. It might not be in the interest of all parties involved to find such an upper ontology and share it. Especially when economic interests are involved, companies might want to keep their developments and achievements hidden. Given that major developments in AI are currently driven by large companies, competitive pressure could overrule scientific interest. Major work being done by researches located in the Western world or China, voices and ideas from other parts may get overheard and no full picture of perception of geographic space can be reached. Use of such models could then lead to an understanding built upon the use itself, as questions must be adapted to answering capabilities. The resulting ontology would be more related to the processing of geographic data than geographic phenomena themselves.

5.3 Future Problems

As mentioned above, building AI models is projected to grow to a trillion dollar market. Developments are largely driven by economic interests and deployable market applications. While science provides foundational mechanisms and ideas, the focus mostly remains on creating models for commercial use. Competition is continuing to grow in the field but policies, increasing costs in infrastructure and talent put a small number of companies at the forefront of new developments. These companies do not work in a transparent way natural to science, but seek to protect their developments and methods. It remains unclear how models are built and how they lead to results. Additionally, tying inquiry to possible marketability runs the risk that the topic might be abandoned if commercial expectations are not met. If the models do not yield the profits hoped for, developments could stop. The lack of transparency and increasing costs tied to building AI models could bar those who would remain interested from working on newer GeoAI foundation models. Another problem is general data quality. The quality of directly sourced data could be increased through new standards and norms, publicly sourced data increasingly suffers in quality as the internet is being flooded with AI-generated content. It is becoming increasingly more difficult to differentiate artificial data from data recorded by humans. With AI models gaining capabilities to create maps or other geographic information, data sources are becoming muddled. Considering the large amount of data needed for a foundation model, this must be avoided at all cost and methods need to be found to make distinctions. Further, as formulating an upper ontology in GIScience led to no results so far, it might be simply not possible to harmonize the widely different definitions and perspectives of geography altogether. If perspectives are too different around the world, it might be unavoidable to make compromises and consider geographic information as inherently fuzzy and imprecise. GeoAI foundation models could lead to acceptable approximations, but could never replace the human perspective of a specific environment. Creating area-specific GeoAI models could help here, but it is questionable whether these could be considered truly foundation models.

6 Conclusion

While several approaches to formulating an upper ontology in GIScience have been made, no clear result could be achieved. Different views on the topic remained until it had been abandoned. Even though treating geographic objects as bona fide is interesting, competing definitions of these objects render the idea of little use. Formalizing an upper ontology based on linguistic descriptions and use seems as a more fruitful approach as reflected in the far larger amount of literature available. This seems more true now than ever before. With new developments in AI and language processing, such upper ontologies could become extremely helpful. As GeoAI foundation model research is a very young area of research, actual implementations are lacking and no clear approach has been formulated. While the topic is of great interest in science and the economy, it is questionable whether it is achievable given the lack of an upper ontology. Interest in GeoAI might revive the formalization of GIScience upper ontologies as they are necessary for successful spatial reasoning in possible foundation models. If interest disappears, so might the idea of GeoAI foundation models. It is too early to make a qualified prediction.

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